

## Drought-induced changes in soil contact and hydraulic conductivity for roots of *Opuntia ficus-indica* with and without rhizosheaths

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### Abstract

Water movement between roots and soil can be limited by incomplete root–soil contact, such as that caused by air gaps due to root shrinkage, and can also be influenced by rhizosheaths, composed of soil particles bound together by root exudates and root hairs. The possible occurrence of air gaps between the roots and the soil and their consequences for the hydraulic conductivity of the root–soil pathway were therefore investigated for the cactus *Opuntia ficus-indica*, which has two distinct root regions: a younger, distal region where rhizosheaths occur, and an older, proximal region where roots are bare. Resin-embedded sections of roots in soil were examined microscopically to determine root–soil contact for container-grown plants kept moist for 21 days, kept moist and vibrated to eliminate air gaps, droughted for 21 days, or droughted and vibrated. During drought, roots shrank radially by 30% and root–soil contact in the bare root region of nonvibrated containers was reduced from 81% to 31%. For the sheathed region, the hydraulic conductivity of the rhizosheath was the least limiting factor and the root hydraulic conductivity was the most limiting; for the bare root region, the hydraulic conductivity of the soil was the least limiting factor and the hydraulic conductivity of the root–soil air gap was the most limiting. The rhizosheath, by virtually eliminating root–soil air gaps, facilitated water uptake in moist soil. In the bare root region, the extremely low hydraulic conductivity of the root–soil air gap during drought helped limit water loss from roots to a drier soil.

### Introduction

Roots must maintain good contact with soil particles for efficient absorption of water and nutrients. When root–soil contact is incomplete, as in extremely porous or fissured soil, or reduced, as by root or soil shrinkage, a plant's nutritional and transpirational demands may not be fully satisfied (Faiz and Weatherley, 1982; Veen et al., 1992). Good root–soil contact is particularly difficult for plants in sandy, shallow, or shifting soils. Not only may such soils be loose in structure, leading to air spaces around roots, but they are also likely to drain readily and hence to dry rapidly. Extreme cases of porous or quickly draining soil occur in deserts and sand dunes, as well as in rock outcroppings or tree canopy sites where soil volume is limited. The tendency of the roots of certain species to form rhizosheaths,

composed of soil particles bound together with root hairs and root exudates, can be advantageous in such habitats; indeed, rhizosheaths develop for roots of dune grasses such as *Oryzopsis hymenoides* and *Agropyron dasystachyum* (Wullstein et al., 1979), the tropical epiphytes *Epiphyllum phyllanthus* and *Rhipsalis baccifera* (North and Nobel, 1994), and the desert cactus *Ferocactus acanthodes* (North and Nobel, 1992).

Roots of the commonly cultivated prickly pear cactus, *Opuntia ficus-indica* (L.) Miller, also form rhizosheaths, which are more tightly bound in dry than in wet soil (Huang et al., 1993; North and Nobel, 1992). As is the case for other species, such as *Zea mays* (McCully and Canny, 1988) and *Triticum aestivum* (Young, 1995), rhizosheaths of *O. ficus-indica* are limited to distal root regions, which generally have a higher water content than do the more proximal regions (Huang et al., 1993; North and Nobel, 1992). Both the

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immaturity of the xylem in the distal root region (Wang et al., 1991) and the properties of the rhizosheath itself may contribute to the greater water content of sheathed roots as opposed to bare roots (Bristow et al., 1985; Huang et al., 1993). The interactions between the rhizosheath, root water status, and root–soil contact can be explored for one-month-old roots of *O. ficus-indica* due to the presence of a sheathed distal region and a bare proximal region along the same root.

In drying soil, or in soil that is locally depleted of water by plant transpiration, root shrinkage may lead to air gaps between the roots and the soil, diminishing the hydraulic conductivity of the root–soil interface (Faiz and Weatherley, 1982; Huck et al., 1970; Nobel and Cui, 1992a, b). The actual decrease in hydraulic conductivity depends on the width of the air gap, the location of the root within the air space, and the number of points along the root where it contacts the soil. Models of water uptake by roots at least partially surrounded by air gaps have been based on the root–soil contact angle (Nye, 1994) and by the fraction of the root periphery in contact with the soil (Fernández and McCree, 1991; Herkelrath et al., 1977; Veen et al., 1992). Another approach, recently applied to the desert succulent *Agave deserti*, is to measure the root–soil gaps in situ for roots in drying soil and to predict the overall hydraulic conductivity for the root–soil system by calculating the conductivity of each component in series: the root, the root–soil air gap, and the soil (North and Nobel, 1997).

Container-grown plants of *Opuntia ficus-indica* were subjected to drought, and root–soil contact was examined for both the sheathed root region and the bare root region, using resin-infiltrated sections of soil containing roots. The water content and the water potential were determined for the soil, the roots, and the rhizosheath; the hydraulic conductivities of all components were calculated to predict the effects of the rhizosheath and the root–soil air gaps on water movement between the roots and the soil. One hypothesis tested was that the rhizosheath facilitates water uptake from a moist soil. A second hypothesis was that both the rhizosheath and the root–soil air gaps help limit root water loss to a drier soil, which would be advantageous for a species such as *O. ficus-indica* that is exposed for long periods to soil water potentials lower than that of its succulent, water-storing shoot.

## Materials and methods

Terminal cladodes (flattened, succulent stem segments) approximately 20 cm long and 12 cm wide were removed from mature plants of *Opuntia ficus-indica* (L.) Miller (Cactaceae) growing in a glasshouse at the University of California, Los Angeles; the basal one-third was placed in cylindrical polystyrene containers 15 cm tall and 10 cm in diameter, which were filled with sandy desert soil sieved to remove particles greater than 3 mm in diameter. The soil was gently tamped to remove air pockets, and the soil line was marked on the container with ink to determine possible changes in soil volume. An aquarium-type airstone had been placed in the soil near the base of each container and fitted to Tygon tubing that extended through a hole drilled in the side of the container, allowing a partial vacuum to be applied to insure complete infiltration of the soil by resin. Plants were maintained in the glasshouse, receiving water twice weekly, with daily maximum/minimum air temperatures averaging 28 °C/16 °C, daily maximum/minimum relative humidities of 70%/40%, and a mean daily photosynthetic photon flux density of 30 mol m<sup>-2</sup> d<sup>-1</sup> (daily maximum instantaneous values of 1500 μmol m<sup>-2</sup> s<sup>-1</sup>). After 10 d, plants had approximately 20 roots averaging 6 cm in length.

At 14 d after planting, ten plants were randomly assigned to each of four groups and maintained for an additional 21 d: 1) control, which was watered twice weekly; 2) vibrated, in which containers were watered twice weekly while placed on a thin aluminum plate that was struck four times with a mallet twice daily (Faiz and Weatherley, 1982; North and Nobel, 1997); 3) droughted by withholding water for the 21 d; and 4) droughted and vibrated. To determine soil bulk density, six containers of soil without plants were also droughted for 21 d, three with vibration treatments and three without. Four additional plants were grown in moist soil, excavated after 14 d for removal of the rhizosheaths, and returned to the soil for 14 d to determine whether rhizosheaths would re-form around existing roots. After the 21 d, five containers from each of the four treatment groups were infiltrated with resin, and the other five were used for determinations of water content and water potential. For the control and vibrated groups, containers had been watered 3 d prior to measurements.

The root water potential ( $\Psi_{\text{root}}$ ) was measured on 10-mm segments approximately centrally located in each of two root zones: the sheathed zone (the dis-

tal 15–200 mm) and the bare zone (the basal 30 mm, which lacked a rhizosheath). In a humidified chamber, the rhizosheath was removed by gently scraping the root surface with a small spatula, and  $\Psi_{\text{root}}$  was determined for previously sheathed roots and bare roots after the segments equilibrated for 2 h in a thermocouple psychrometer (Tru Psi, Decagon Devices, Pullman, WA). The water potential of the removed rhizosheaths ( $\Psi_{\text{sheath}}$ ) and the shoot water potential ( $\Psi_{\text{cladode}}$ ) for 9-mm diameter cores taken from mid-cladode were also measured psychrometrically. Water content was determined for sheathed roots with rhizosheaths attached, sheathed roots with rhizosheaths removed, bare roots, and shoot segments by obtaining the fresh weights of samples and their dry weights after 48 h at 70 °C. For both soil and plant material, water content was calculated as (fresh weight – dry weight) / fresh weight).  $\Psi_{\text{soil}}$  was routinely determined gravimetrically using a moisture-release curve for the desert soil that was used in the containers (Young and Nobel, 1986) and the measured bulk density of the soil for vibrated and nonvibrated containers; such determinations of  $\Psi_{\text{soil}}$  were within  $\pm 7\%$  of values obtained with the thermocouple psychrometer ( $n = 4$  containers for moist and droughted conditions).

### Resin infiltration and sectioning

An acrylic resin that hardens under moist as well as dry conditions was used to make sections of roots in soil in the containers (Moran et al., 1989). For each batch, 34 g of Araldite GY 509 resin, 34 g of Araldite RD-2 diluent, and 32 g of HY 956 hardener (Ciba-Geigy, Hawthorne, NY) were used. Three batches of liquid resin were applied in sequence from a 60-mL syringe fitted with a pipette tip and suspended 2 cm above the soil surface, which dispensed the resin at approximately  $50 \mu\text{L s}^{-1}$  (total time 1.5 h). For plants in containers, a partial vacuum was applied during the application of the third resin batch to improve infiltration into the soil. One h after infiltration, the resin was sufficiently hardened to allow removal of the soil block, and the block was further infiltrated with resin under partial vacuum in the laboratory. At 24–48 h, the blocks were trimmed and cut in 1-cm-thick transverse and longitudinal sections with a diamond lapidary saw. Sections were viewed and photographed under a dissecting microscope at magnifications of 10–30 $\times$ . The soil plus roots from one container that had been droughted for 21 d was cut into 10-cm blocks, which were wrapped in aluminum foil and frozen in liquid nitrogen; these

blocks were subsequently vacuum-dried in a lyophilizer (Lyph-Lock 4.5, Labconco, Kansas City, MO) prior to resin infiltration as above. Sections 50 to 100  $\mu\text{m}$  in thickness were cut with the lapidary saw, mounted on slides, and viewed with a compound microscope (magnifications of 30–100 $\times$ ), using polarized light.

Measurements of root–soil contact, root diameters, and widths of root–soil air gaps were made from sections examined with the dissecting microscope using an ocular micrometer and from photographs enlarged to final magnifications of 15–60 $\times$ . An air gap width was the mean of four measurements at 90° intervals around the root periphery. Only roots cut in cross-section were considered. The percentage of root–soil contact was determined by outlining the root perimeter in the photographs with a thread on which regions with no gap were marked in ink. Using the dissecting microscope, rhizosheath thicknesses were determined by taking the mean of four measured diameters for each sheathed root segment, then removing the rhizosheath and measuring the diameter of the unsheathed root; the rhizosheath thickness was calculated as one half of the difference between the sheathed and the unsheathed root segment. Data were analyzed using one way ANOVA followed by Student-Newman-Keuls pairwise testing; in two cases with non-normal distribution or unequal variances, the non-parametric Kruskal-Wallis test with pairwise testing by Dunn's method was used (SigmaStat, Jandel Scientific, San Rafael, CA).

### Hydraulic conductivity

For sheathed root regions, the hydraulic conductivity ( $\text{m s}^{-1}\text{MPa}^{-1}$ ) of the overall root–soil pathway ( $L_{\text{overall}}$ ) for water movement was based on three components – the soil, the rhizosheath, and the root:

$$\frac{1}{L_{\text{overall}}} = \frac{1}{L_{\text{soil}}^{\text{eff}}} + \frac{1}{L_{\text{sheath}}^{\text{eff}}} + \frac{1}{L_{\text{P}}} \quad (1)$$

where  $L_{\text{soil}}^{\text{eff}}$  is the effective hydraulic conductivity of the soil,  $L_{\text{sheath}}^{\text{eff}}$  is the effective hydraulic conductivity of the rhizosheath, and  $L_{\text{P}}$  is the hydraulic conductivity of the root (Figure 1A).  $L_{\text{soil}}^{\text{eff}}$  was calculated as follows (Nobel and Cui, 1992a):

$$L_{\text{soil}}^{\text{eff}} = \frac{L_{\text{soil}}}{r_{\text{root}} \ln(r_{\text{distant}}/r_{\text{sheath}})} \quad (2)$$

where  $L_{\text{soil}}$  is the soil hydraulic conductivity coefficient ( $\text{m}^2 \text{s}^{-1} \text{MPa}^{-1}$ ), which is a function of  $\Psi_{\text{soil}}$ , as has

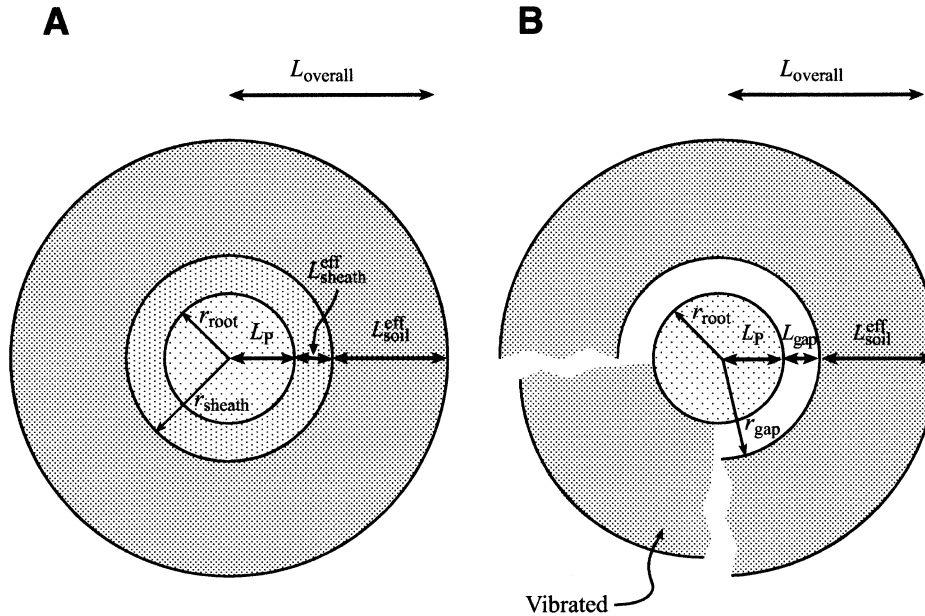


Figure 1. Cross-sections of (A) a sheathed root (light stippling) surrounded by a rhizosheath (medium stippling), with the bulk soil (heavy stippling) to the outside; and (B) a bare root after shrinkage during drought, surrounded by an air gap (no stippling) and the bulk soil. Radii of the root ( $r_{\text{root}}$ ), the air gap ( $r_{\text{gap}}$ ), and the rhizosheath ( $r_{\text{sheath}}$ ) are indicated. Arrows indicate the components of the overall hydraulic conductivity ( $L_{\text{overall}}$ ): the effective soil hydraulic conductivity ( $L_{\text{soil}}^{\text{eff}}$ ), the root hydraulic conductivity ( $L_p$ ), the effective hydraulic conductivity of the rhizosheath ( $L_{\text{sheath}}^{\text{eff}}$ ; A only) and the hydraulic conductivity of the air gap ( $L_{\text{gap}}$ ; B only). For B, lower left quadrant indicates the effect of the vibration treatment.

been determined for the desert soil used in the containers (Young and Nobel, 1986);  $r_{\text{sheath}}$  and  $r_{\text{root}}$  (m) are the radii of the rhizosheath and root, respectively (Figure 1A); and  $r_{\text{distant}}$  is set to half of the inter-root spacing (Caldwell, 1976; Nobel and Cui, 1992a), which was about 15 mm for the plants considered, whose roots were fairly evenly distributed in the containers.  $L_{\text{sheath}}^{\text{eff}}$  was calculated from the gravimetric moisture content of the rhizosheath and the radii of the root and the rhizosheath, assuming the sheath soil to have the same moisture-release characteristics as the desert soil.  $L_p$  was based on previous measurements for young main roots of *O. ficus-indica* (North and Nobel, 1992). For bare root regions,  $L_{\text{overall}}$  included the hydraulic conductivity of a possible air gap between the root and the soil,  $L_{\text{gap}}$ , in place of  $L_{\text{sheath}}^{\text{eff}}$  (Figure 1B):

$$\frac{1}{L_{\text{overall}}} = \frac{1}{L_{\text{soil}}^{\text{eff}}} + \frac{1}{L_{\text{gap}}} + \frac{1}{L_p} \quad (3)$$

The hydraulic conductivity of the air gap,  $L_{\text{gap}}$ , was calculated assuming isothermal conditions and a concentric position of the root within the air space (Nobel

and Cui, 1992a):

$$L_{\text{gap}} = \frac{L'}{r_{\text{root}} \ln(r_{\text{gap}}/r_{\text{root}})} \quad (4)$$

where  $r_{\text{gap}}$  equals the radius of the air space;  $L'$  equals  $\bar{V}_w^2 D_{wv} P_{wv}^* / (RT)^2$ , where  $\bar{V}_w$  is the partial molal volume of water ( $\text{m}^3 \text{mol}^{-1}$ );  $D_{wv}$  is the diffusion coefficient of water vapour in air ( $\text{m}^2 \text{s}^{-1}$ );  $P_{wv}^*$  is the saturation partial pressure of water (MPa); and  $RT$  is the gas constant times the absolute temperature ( $\text{m}^3 \text{MPa mol}^{-1}$ ). At 25°C,  $L'$  equals  $4.18 \times 10^{-12} \text{ m}^2 \text{s}^{-1} \text{MPa}^{-1}$  (Nobel and Cui, 1992a).

## Results

### Dimensions of roots, rhizosheaths, and root–soil contact

The distal 150- to 200-mm regions of roots of *O. ficus-indica* were surrounded by rhizosheaths composed of root hairs and soil particles that adhered loosely to roots from moist soil (Figure 2A). Rhizosheaths did not reform after they had been removed from roots that were

**Table 1.** Dimensions of the rhizosheath, root, and root–soil air gap for *Opuntia ficus-indica*. Root diameters were measured in the sheathed root region (50 mm from the root tip) and the bare root region (20 mm from the cladode base). Rhizosheath thickness was calculated as half the difference between the diameter of the root plus rhizosheath minus the diameter of the root after the rhizosheath was removed. The width of the air gap was the mean of four measurements from the outer edge of the root in transverse section to the edge of the soil (determined for the bare root region only). The percentage of root–soil contact was determined from photographs by outlining the perimeter of the root with thread and marking the thread for regions where no gap was present. Data are means  $\pm$  s.e. for  $n = 12$  roots

| Treatment              | Rhizosheath thickness (mm) | Sheathed region root radius (mm) | Bare region root radius (mm) | Root–soil air gap width (mm) | Root–soil contact (%)       |
|------------------------|----------------------------|----------------------------------|------------------------------|------------------------------|-----------------------------|
| Control                | 0.80 $\pm$ 0.06            | 0.81 $\pm$ 0.05                  | 0.89 $\pm$ 0.05              | 0.13 $\pm$ 0.02              | 81.4 $\pm$ 4.6              |
| Vibrated               | 0.74 $\pm$ 0.07            | 0.89 $\pm$ 0.06                  | 0.82 $\pm$ 0.05              | 0.06 $\pm$ 0.03              | 87.6 $\pm$ 9.1              |
| Droughted              | 0.96 $\pm$ 0.08            | 0.66 $\pm$ 0.04 <sup>a</sup>     | 0.66 $\pm$ 0.03 <sup>a</sup> | 0.44 $\pm$ 0.05 <sup>a</sup> | 30.5 $\pm$ 6.3 <sup>a</sup> |
| Droughted/<br>vibrated | 0.91 $\pm$ 0.07            | 0.54 $\pm$ 0.05 <sup>b</sup>     | 0.61 $\pm$ 0.05 <sup>a</sup> | 0.10 $\pm$ 0.02              | 81.0 $\pm$ 3.1              |

<sup>a,b</sup> Different superscripts within a column indicate significant differences from the control and between treatments ( $p < 0.05$  for pairwise testing).

subsequently replanted; i.e., rhizosheaths formed only once for each root in the region of new growth. Vibration treatments (to eliminate root–soil air gaps) had no effect on the thickness of the rhizosheath, which nearly equalled the radius of roots in moist soil (Table 1). Soil particles adhered more tightly to roots under droughted than under moist conditions, and rhizosheaths tended to be thicker (but  $p > 0.05$ ). In the sheathed root region of droughted roots, the rhizosheath thickness exceeded the root radius, and the root radius for both nonvibrated and vibrated containers was about 30% smaller than that of the control under moist conditions ( $p < 0.05$ ; Table 1). Mean diameters of roots in resin sections were similar to those of freshly excavated roots ( $p > 0.05$ ;  $n = 8$ ). Older root regions (extending 20–30 mm from the base of a cladode) lacked rhizosheaths and had extensive secondary xylem in the vascular cylinder (Figure 2B). The radius of bare roots also decreased by about 30% during drought ( $p < 0.05$ ; Table 1).

Rhizosheaths could not be distinguished from the bulk soil in resin-infiltrated soil sections (Figure 2C). Control roots (from moist soil) in the sheathed root region were white and rounded in cross-section, with a small central vascular cylinder. After 21 d of drought, sheathed roots were slightly less round in cross-section, and the vascular cylinder was more prominent (Figure 2C). Under both control and droughted conditions, roots from the bare root region were also somewhat angular, with a well-developed vascular cylinder (Figure 2D). Roots from soil sections that were freeze-dried prior to resin infiltration were less round and showed

greater cellular distortion than roots embedded without freeze-drying.

In the bare root region, small air gaps were present between the root and moist soil, and the root–soil contact averaged 81% of the root periphery (Table 1). Vibrating the containers had no significant effect on gap width or root–soil contact for such roots. After 21 d of drought, the mean width of the root–soil air gap in the bare root region was three times greater than under control conditions ( $p < 0.05$ ; Table 1; Figure 2D, E); vibrating the containers during drought resulted in a gap width similar to that of the control (Table 1; Figure 2F). Root–soil contact in the bare root region also decreased during drought ( $p < 0.05$ ), except for roots from vibrated containers (Table 1).

#### *Water relations of the soil, rhizosheath, and organs of O. ficus-indica*

The bulk density of the desert soil used did not change significantly due to drought or vibration treatments (Table 2). The water content of the soil decreased by a factor of three during drought, leading to a decrease in the soil water potential ( $\Psi_{\text{soil}}$ ) from  $-0.09$  MPa to an average of  $-2.7$  MPa. Vibration treatments had no effect on soil water content or  $\Psi_{\text{soil}}$  for either moist or droughted conditions (Table 2). Under moist soil conditions, vibration also had no effect on the water content or water potential of the rhizosheath ( $\Psi_{\text{sheath}}$ ). The water content of the rhizosheath decreased by approximately a factor of two during drought for both vibrated

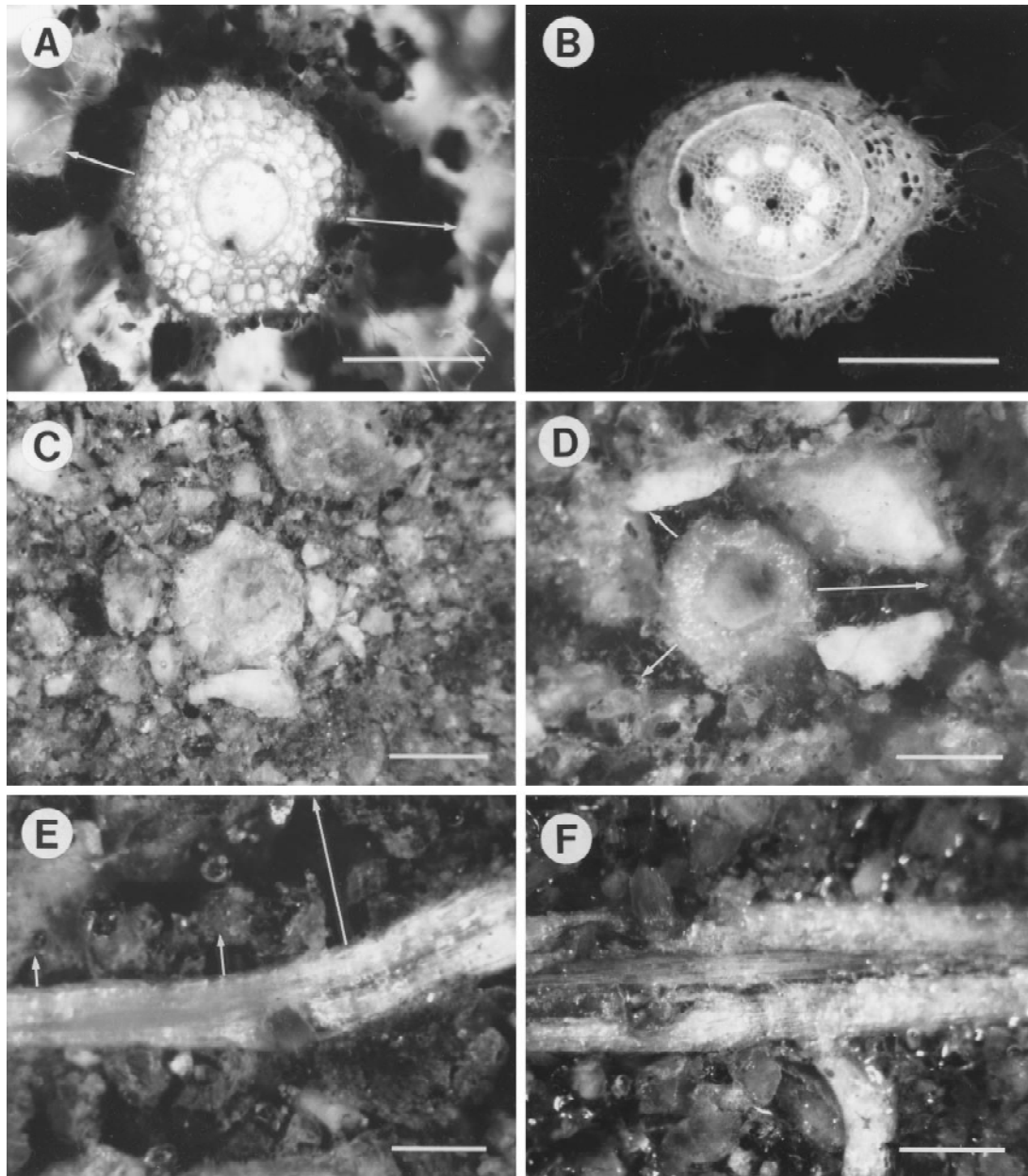


Figure 2. Micrographs of roots of *Opuntia ficus-indica*: fresh cross-sections (viewed using polarized light) of (A) a distal, sheathed root from moist soil with rhizosheath indicated by arrows and (B) a bare root lacking a rhizosheath; resin-embedded cross-sections of (C) a sheathed root and (D) a bare root in soil after 21 d of drought; and resin-embedded longitudinal sections of bare roots after drought from (E) nonvibrated and (F) vibrated containers. Arrows indicate air gaps; scale bars equal 1.0 mm.

and nonvibrated containers;  $\Psi_{\text{sheath}}$  decreased as well during drought ( $p < 0.05$ ; Table 2).

The water content of distal root segments from the sheathed root region (with rhizosheaths removed just prior to measurement) was unaffected by vibration

under moist conditions (Table 3). During drought, the water content of segments from the sheathed region decreased by 29% and 35% for the nonvibrated and vibrated containers, respectively ( $p < 0.05$ ), and the water potential of the root segments ( $\Psi_{\text{root}}$ ) decreased

Table 2. Soil and rhizosheath water relations parameters for *O. ficus-indica*. Soil properties were determined for soil from mid-container in the middle of the root zone; data are means  $\pm$  s.e. for  $n = 8$  plants

| Treatment              | Soil                                   |                      |                          | Rhizosheath          |                          |
|------------------------|----------------------------------------|----------------------|--------------------------|----------------------|--------------------------|
|                        | Bulk density<br>( $\text{Mg m}^{-3}$ ) | Water content<br>(%) | Water potential<br>(MPa) | Water content<br>(%) | Water potential<br>(MPa) |
| Control                | $1.23 \pm 0.10$                        | $9.4 \pm 0.9$        | $-0.09 \pm 0.08$         | $10.5 \pm 0.9$       | $-0.08 \pm 0.08$         |
| Vibrated               | $1.32 \pm 0.05$                        | $8.8 \pm 1.0$        | $-0.09 \pm 0.03$         | $11.8 \pm 1.3$       | $-0.05 \pm 0.02$         |
| Droughted              | $1.33 \pm 0.11$                        | $2.8 \pm 0.2^a$      | $-2.26 \pm 0.39^a$       | $4.5 \pm 0.2^a$      | $-1.03 \pm 0.09^a$       |
| Droughted/<br>vibrated | $1.37 \pm 0.13$                        | $2.5 \pm 0.4^a$      | $-3.08 \pm 0.60^a$       | $4.2 \pm 0.3^a$      | $-1.17 \pm 0.10^a$       |

<sup>a</sup>Superscripts within a column indicate significant differences from the control ( $p < 0.05$  for pairwise testing).

Table 3. Water content and water potential for root segments from sheathed root regions (with rhizosheaths removed), root segments from bare root regions, and cladodes of *O. ficus-indica*. Data are means  $\pm$  s.e. for  $n = 8$  plants

| Treatment             | Water content (%) |                  |                  | Water potential (MPa) |                    |                    |
|-----------------------|-------------------|------------------|------------------|-----------------------|--------------------|--------------------|
|                       | Sheathed region   | Bare region      | Cladode          | Sheathed region       | Bare region        | Cladode            |
| Control               | $90.1 \pm 0.9$    | $71.4 \pm 0.9$   | $91.6 \pm 0.3$   | $-0.41 \pm 0.05$      | $-0.76 \pm 0.04$   | $-0.69 \pm 0.05$   |
| Vibrated              | $90.3 \pm 1.2$    | $76.2 \pm 1.1^a$ | $91.5 \pm 1.3$   | $-0.33 \pm 0.05$      | $-0.74 \pm 0.06$   | $-0.72 \pm 0.04$   |
| Droughted             | $64.1 \pm 2.2^a$  | $61.2 \pm 1.4^b$ | $85.6 \pm 1.3^a$ | $-1.05 \pm 0.06^a$    | $-0.98 \pm 0.03^a$ | $-0.84 \pm 0.02^a$ |
| Droughted<br>vibrated | $58.6 \pm 2.0^b$  | $57.8 \pm 1.7^b$ | $86.0 \pm 0.8^a$ | $-0.99 \pm 0.04^a$    | $-1.18 \pm 0.06^b$ | $-0.83 \pm 0.03^a$ |

<sup>a,b</sup>Different superscripts within a column indicate significant differences from the control and between treatments ( $p < 0.05$  for pairwise testing).

by approximately 0.6 MPa ( $p < 0.05$ ; Table 3). For segments from the proximal, bare root region under moist soil conditions, the water content was higher for vibrated than for control containers ( $p < 0.05$ ), although  $\Psi_{\text{root}}$  was similar (Table 3). During drought, the water content of segments from the bare root region decreased by 14% and 25% for nonvibrated and vibrated containers, respectively ( $p < 0.05$ );  $\Psi_{\text{root}}$  decreased as well, with a greater decrease for bare roots from the vibrated containers ( $p < 0.05$ ; Table 3). The water potential of bare roots from droughted, vibrated containers (lacking root–soil air gaps) was also lower than the pooled  $\Psi_{\text{root}}$  for sheathed roots from vibrated and nonvibrated containers and for bare nonvibrated roots ( $p < 0.05$ ).

For cladodes, both the water content and the water potential ( $\Psi_{\text{cladode}}$ ) decreased during drought ( $p < 0.05$ ) and were unaffected by vibration treatments (Table 3). Under moist conditions,  $\Psi_{\text{cladode}}$  was lower than  $\Psi_{\text{root}}$  for the sheathed root region for both vibrated and nonvibrated containers ( $p < 0.01$ ). For droughted containers,  $\Psi_{\text{cladode}}$  was higher than  $\Psi_{\text{root}}$  in the sheathed region for both nonvibrated and vibrated containers ( $p < 0.01$ ). For the bare root region under moist condi-

tions,  $\Psi_{\text{cladode}}$  was similar to  $\Psi_{\text{root}}$  ( $p > 0.05$ ), yet under droughted conditions,  $\Psi_{\text{cladode}}$  was greater than  $\Psi_{\text{root}}$  ( $p < 0.01$ ; Table 3).

#### Hydraulic conductivity of the root–soil pathway

For all treatments, the hydraulic conductivity of the root–soil pathway ( $L_{\text{overall}}$ ) in the sheathed root region was most limited by the root hydraulic conductivity ( $L_p$ ) and least limited by the conductivity of the rhizosheath ( $L_{\text{sheath}}^{\text{eff}}$ ; Table 4). Drought caused both  $L_{\text{sheath}}^{\text{eff}}$  and the hydraulic conductivity of the soil ( $L_{\text{soil}}^{\text{eff}}$ ) to decrease substantially, and  $L_{\text{overall}}$  for the sheathed region was approximately 50% lower than under moist conditions. Under both moist and droughted conditions,  $L_{\text{overall}}$  for the sheathed region was higher than that for the bare root region (Table 4). In the bare root region, the hydraulic conductivity of the root–soil air gap ( $L_{\text{gap}}$ ) limited  $L_{\text{overall}}$  for all treatments and was lowest for droughted, nonvibrated containers;  $L_{\text{overall}}$  for droughted and vibrated containers was similar to that for the control, as the increase in  $L_{\text{gap}}$  due to the vibration treatment largely compensated for the decreases in  $L_{\text{soil}}^{\text{eff}}$  and  $L_p$  due to drought (Table 4).

Table 4. Hydraulic conductivities of the soil ( $L_{\text{soil}}^{\text{eff}}$ ), the rhizosheath ( $L_{\text{sheath}}^{\text{eff}}$ ) for the sheathed root region, the root–soil air gap ( $L_{\text{gap}}$ ) for the bare root region, the root ( $L_p$ ), and the overall root–soil pathway ( $L_{\text{overall}}$ ). Data were calculated using root, rhizosheath, and gap dimensions (Table 1), water potentials of the soil and rhizosheath (Table 2), and  $L_p$  for wet and droughted sheathed roots and bare roots of *O. ficus-indica* (North and Nobel, 1992) in Equations (1–4), assuming isothermal conditions and concentric location of the roots within air gaps

| Treatment              | Sheathed root region                          |                                  |       |                      | Bare root region                              |                  |       |                      |
|------------------------|-----------------------------------------------|----------------------------------|-------|----------------------|-----------------------------------------------|------------------|-------|----------------------|
|                        | $L_{\text{soil}}^{\text{eff}}$                | $L_{\text{sheath}}^{\text{eff}}$ | $L_p$ | $L_{\text{overall}}$ | $L_{\text{soil}}^{\text{eff}}$                | $L_{\text{gap}}$ | $L_p$ | $L_{\text{overall}}$ |
|                        | $(10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1})$ |                                  |       |                      | $(10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1})$ |                  |       |                      |
| Control                | 430                                           | 2050                             | 1.3   | 1.30                 | 356                                           | 0.35             | 2.7   | 0.31                 |
| Vibrated               | 390                                           | 4420                             | 1.3   | 1.30                 | 332                                           | 0.72             | 2.7   | 0.57                 |
| Droughted              | 2.39                                          | 19.5                             | 1.0   | 0.68                 | 2.01                                          | 0.12             | 0.9   | 0.10                 |
| Droughted/<br>vibrated | 1.37                                          | 14.7                             | 1.0   | 0.56                 | 1.41                                          | 0.45             | 0.9   | 0.25                 |

## Discussion

Roots of *Opuntia ficus-indica* shrank radially by about 30% during drought in both the distal root region, where rhizosheaths were present, and in the proximal root region, where roots were bare. A radial shrinkage of 40% has been observed for young roots of *O. ficus-indica* that were exposed to a more desiccating vapour phase (water potential of 10 MPa; Nobel and Cui, 1992a). Resin-infiltrated soil sections of roots in situ indicated that root shrinkage was accompanied by the development of root–soil air gaps in the bare root region but not in the sheathed root region. Smaller root–soil air gaps also occurred in the bare root region under moist soil conditions, resulting in an average root–soil contact of 81%. Some loss of contact with the soil for the older root regions may have been due to the death and subsequent shrinkage of epidermal and cortical cells that occur as roots of *O. ficus-indica* age (North and Nobel, 1996). In addition, the maturation of metaxylem vessels and the development of secondary xylem in the proximal root region (North and Nobel, 1992) increase the rate of longitudinal water transport, which is accompanied by increased water loss from the outer root tissues as well as the disappearance of the rhizosheath (Wang et al., 1991). In contrast to the bare roots, sheathed roots of *O. ficus-indica* apparently maintained essentially full contact with the soil, even under droughted conditions.

Under moist conditions, both the water content and the water potential ( $\Psi_{\text{root}}$ ) were higher for the sheathed region than for the bare root region of *O. ficus-indica*, as is also the case for roots of *Zea mays* (Wang et al., 1991). The contribution of the rhizosheath to the

enhanced water status of the sheathed roots was apparently not due to the water relations of the rhizosheath itself, at least under moist conditions, as its water content and water potential did not differ from those of the bulk soil. In addition, the predicted hydraulic conductivity of the root–soil pathway ( $L_{\text{overall}}$ ) was not changed when calculated either without the hydraulic conductivity of the rhizosheath ( $L_{\text{sheath}}^{\text{eff}}$ ) or with the conductivity of an equal thickness of the bulk soil substituted for  $L_{\text{sheath}}^{\text{eff}}$  in Equation (1). Under both moist and droughted conditions, the overriding limitation on  $L_{\text{overall}}$  in the sheathed root region was the hydraulic conductivity of the root ( $L_p$ ), rendering changes in the much greater  $L_{\text{sheath}}^{\text{eff}}$  inconsequential. The rhizosheath had an indirect effect on  $L_{\text{overall}}$ , however, by helping to maintain full contact between sheathed roots and the soil. In the bare root region, the hydraulic conductivity of the root–soil air gap ( $L_{\text{gap}}$ ) was the lowest and hence the most limiting component of  $L_{\text{overall}}$ ; thus the rhizosheath, by eliminating or reducing root–soil air gaps, facilitated water movement across the root–soil interface in the sheathed root region. Rhizosheaths also enhance water uptake for the dune grass *Oryzopsis hymenoides* according to a numerical simulation for sheathed versus bare roots (Bristow et al., 1985).

Under droughted conditions, the water content and the water potential of the rhizosheath ( $\Psi_{\text{sheath}}$ ) exceeded those of the bulk soil. The higher water content of the rhizosheath may have been due to an efflux of water from the roots (McCully, 1995), which shrank radially and decreased in both water content and  $\Psi_{\text{root}}$  during drought. Rhizosheaths of both *Triticum aestivum* (Young, 1995) and *Oryzopsis hymenoides* (Bristow et al., 1985) also have higher water contents than the



bulk soil under a range of soil moisture conditions. After 21 d in drying soil, the difference between  $\Psi_{\text{root}}$  and  $\Psi_{\text{sheath}}$  for sheathed roots of *O. ficus-indica* was not significant, indicating that under droughted conditions the local driving force for water movement at the root surface in the sheathed root region was minimal.

For the bare root region, the hydraulic conductivity of the gap limited the overall hydraulic conductivity. Moreover,  $L_{\text{gap}}$  decreased by a factor of three during drought, as the mean gap width increased threefold and root–soil contact decreased from 81% to 31%.  $L_{\text{overall}}$  also decreased by a factor of three during drought to  $0.10 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$ . Multiplying  $L_P$  by the percent root–soil contact (Veen et al., 1992) and substituting this quantity for  $L_P$  and  $L_{\text{gap}}$  in Equation (3) results in an  $L_{\text{overall}}$  of  $0.24 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$ ; an even closer match results from multiplying  $L_P$  by a "root–soil contact factor" obtained by dividing the volumetric water content of the soil by the saturated soil water content (Herkelrath et al., 1977), giving  $0.14 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$  for  $L_{\text{overall}}$ . Despite the relatively good agreement in values obtained by the three methods for  $L_{\text{overall}}$  under droughted conditions,  $L_{\text{overall}}$  under moist conditions as calculated using Equation (3) was about five to seven times lower than  $L_{\text{overall}}$  calculated on the basis of root–soil contact. Equation (4) used to calculate  $L_{\text{gap}}$  assumes that the root is concentrically located within the air space, whereas most roots, including those of *O. ficus-indica*, touch the soil at one or more points on the root periphery under moist conditions (Kooistra et al., 1992). When a root is placed within an air space simulated by a cylinder of moistened filter paper such that it touches the paper in a line along the cylinder, the measured  $L_{\text{gap}}$  is 2.4-fold greater than for roots concentrically located (Nobel and Cui, 1992b). Even when  $L_{\text{gap}}$  for droughted bare roots of *O. ficus-indica* is multiplied by 2.4, it is still the limiting conductivity in the root–soil pathway. Equation (4) assumes that water crosses the gap as a vapour; however, under moist conditions when a root is in at least partial contact with the soil, the movement of water as a liquid predominates (Nye, 1994). In this case, predictions of  $L_{\text{overall}}$  based on a root–soil contact factor may be more realistic than one that incorporates  $L_{\text{gap}}$ .

Regardless of the exact nature of  $L_{\text{gap}}$ , the root–soil air gap substantially affected water exchange at the root–soil interface during drought, as demonstrated by a number of differences in the bare root region of *O. ficus-indica* between vibrated and nonvibrated containers. The threefold increase in mean gap width that developed for bare roots during drought did not

occur for vibrated containers, and root–soil contact for droughted, vibrated containers was similar to that under moist conditions, in contrast to the 31% root–soil contact for droughted, nonvibrated containers. Also,  $\Psi_{\text{root}}$  was higher for bare roots in droughted, nonvibrated containers, as occurs for nonvibrated versus vibrated roots of the desert succulent *Agave deserti* (North and Nobel, 1997), indicating that the root–soil air gap apparently reduced water loss from the bare root. Furthermore,  $\Psi_{\text{root}}$  for bare roots from droughted, nonvibrated containers did not differ from that for sheathed roots, unlike the case for roots from vibrated containers (lacking root–soil air gaps), in which water potential differences could have led to water movement from the sheathed region to the bare root region and thence to the drier soil. Such reverse water flow (Caldwell and Richards, 1989) would be diminished for bare roots of *O. ficus-indica* that are surrounded by air gaps. A decrease in hydraulic conductivity at the root–soil interface is particularly important for a succulent such as *O. ficus-indica*, in which water can move from upper to lower organs along a water potential gradient (Wang et al., 1997). In any case, despite the decrease in water potential from the cladode to the root to the soil, water loss is low in the bare root region because of the low conductivity of the root–soil air gap.

In conclusion, the rhizosheath facilitated water uptake in the sheathed root region of *O. ficus-indica* under moist conditions by improving root–soil contact. Under droughted conditions, the water potential of the rhizosheath was higher than that of the bulk soil and similar to that of the root, thus minimizing the driving force for water loss from the root surface of sheathed roots. In the bare root region, root–soil air gaps were present in moist soil and increased threefold in width during drought, greatly reducing the hydraulic conductivity of the root–soil interface and limiting root water loss to the drier soil.

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